# ENGR489 Demo Cheat Sheet

Oct 21, 2026

## 0. Setup

Every plot runs from `demo_plots.m` in `FieldTiltScan`, one section at a time: click inside a section and press Cmd+Enter (Ctrl+Enter on Windows). Run section 0 first; never press the green Run button.

In Terminal, get the latest files (only these three are replaced; your scans in `data/` are untouched):

```
cd /Users/rochelle/FInal
git fetch origin claude/upbeat-planck-83lcp0
git checkout origin/claude/upbeat-planck-83lcp0 -- FieldTiltScan/demo_plots.m
FieldTiltScan/DEMO_CHEATSHEET.md FieldPlots/compare_heatmaps.m
```

In MATLAB:

```
clear functions
cd /Users/rochelle/FInal/FieldTiltScan
edit demo_plots.m
```

Section 0 itself (what it runs):

```
assert(isfile("scan_config.m") && isfolder("data"), "cd into FieldTiltScan
first");
addpath("../FieldPlots")
close all
full = "data/full25-4z_corr-hybrid_under_combined_20261003.csv";
bg   = "data/full25-4z_corr-hybrid_none_20261004_155505.csv";
```

Keep `tilt_correct.m`, `compare_runs.m` and `fit_dipole.m` open in other tabs to show the maths if asked.

## 1. Full-volume field map

The magnet's field over 260 points (5 × 13 × 4, z −700 to −600 mm), with the no-magnet background subtracted and tilt corrected.

```
plot_field_layers("data/full25-4z_corr-hybrid_under_combined_20261003.csv",
...
    "data/full25-4z_corr-hybrid_none_20261004_155505.csv", ...
    "Figures", {"heat","3d"}, "TiltCorrect", true, "Labels", false)
```

- **Heat map:** |B| per layer on one colour scale. Peak 0.535 G at (0, −25, −700), above the magnet. Median per layer falls 0.39 → 0.27 → 0.20 → 0.14 G going up.
- **3D map:** the layers stacked at their real heights, arrows = field direction. Rotate it with the mouse.
- **Say:** a dipole falls as $1/r^3$; the top layer is about 380 mm from the magnet against 280 mm for the bottom, so theory gives $(380/280)^3 \approx 2.5\times$ and the data gives 2.7×.
- **Caveat:** the background was taken the next day; drift is about 11 mG/h.

If asked (sections 1b and 1c):

```
plot_field_layers(full, bg, "Figures", {"direction","side"}, "TiltCorrect",
true)
plot_field_arrows3d(full, bg, "TiltCorrect", true)
```

Direction map: arrows = field direction in the x–y plane, colour = Bz. 3D arrows: length = field strength, shorter going up.

## 2. Probe tilt

The platform tilts up to 2.55° (median 1.2°), which shifts the measured field by up to 41 mG, so tilt is logged and corrected.

```
plot_tilt(full);
figure("Name", "Tilt correction change");
image(imread("figures/magnet_tilt_20261002_223300/correction_change.png"));
axis image off
```

- **Figure 1, tilt map (show this):** colour = tilt vs the grid centre, arrows = which way it dips. Smooth and the same on every layer, so it is systematic (geometry, rod-end play, servo calibration), not noise.
- **Figures 2 and 3:** skip. Figure 2's settle-time panel only shows the fixed 5 s wait.
- **Correction-change image:** tilt, |ΔB| and the change in Bx, By, Bz per layer. Mostly Bx and By, because tilting a mostly vertical field swings it sideways.
- **Say:** for small angles $|\Delta B| \approx \theta|B|$ ($\theta$ in radians): 0.045 rad × 0.9 G ≈ 40 mG, about 20× the 2 mG run-to-run spread.

Small-angle rule: $|\Delta B| \approx \theta \times |B|$, with $\theta$ in radians (1° = 0.0175 rad).

## 3. Repeatability (6 Oct, five runs)

Five scans back to back with the magnet fixed changed by 1.4–1.9 mG median per point, which is 2.2 mm median position repeatability.

```
compare_repeat_heatmaps("Units", "mG", "Labels", false);
```

- **Figure 1, heat maps:** the no-magnet reference (raw, about 0.70 G), then runs 1–5 with the magnet only (about 0.12 G). All five look the same.
- **Figure 2, change:** run 1→2 … 4→5 in mG; red = stronger, blue = weaker, white = no change. Largest anywhere 11.7 mG.
- **Figure 3, summary:** mean, mean size and RMS of the change per pair; the same table prints in the command window.
- **Say:** sensor noise is about 0.2 mG, the run-to-run spread about 2 mG, so the spread comes from positioning and settling, not the sensor.

Position repeatability: $\sigma_{pos} = \sigma_B / |\nabla B|$ (field spread across runs ÷ how fast the field changes with position).

If asked (sections 3b and 3c):

```
repeatability_heatmaps("Pairs", [1 2; 1 3; 1 4; 1 5]);
R = compare_runs("data/repeat_under-run*.csv",
"data/repeat_none_20261006_192127.csv", "Plot", false);
plot_repeatability_layers(R, "Title", "5 runs, magnet fixed")
```

- **3b, every run against run 1:** the change grows 1.7 % → 3.1 % and shifts colour across the whole grid: drift of about 11 mG over the hour, about 17 % of it from tilt creep, the rest most likely thermal.
- **3c, position repeatability:** $\sigma_{pos}$ per point (median 2.2 mm; grey × = gradient too weak to convert) and the gradient (median about 116 nT/mm, strongest on the bottom layer).

## 4. Dipole accuracy (saved image)

Moving the magnet a known distance, the joint dipole fit finds the y moves within 1.3 mm of the ruler over 100 and 200 mm.

```
figure("Name", "Dipole joint fit");
image(imread("data/dipole_joint_fit.png")); axis image off
```

- **Top row:** the magnet's field on the bottom layer for each of the five positions; × = fitted, ○ = ruler.
- **Bottom left:** top view of all five fitted positions against the ruler.

- **Bottom right:** each move's fitted length against the ruler.
- **x moves:** off by −10 and +6 mm. The grid is only 100 mm wide in x (300 mm in y), so x is about 5.5 mm 1σ on a shift; these are 1–2σ.
- **Model:** one moment m shared by all five runs, one position per run. The single-run fit ( `fit_dipole.m` ) gives 14–17 mm errors because tilt of m and position trade off; sharing m removes that.

Dipole model: $B(r) = (\mu_0 / 4\pi) \times [3(m\cdot\hat{u})\hat{u} - m] / |r|^3$, where $\hat{u} = r / |r|$.

## 5. Live scan

A 9-point live scan takes about 2 min; close the delta app first (only one program can use the serial port), then run from Terminal:

```
cd /Users/rochelle/FInal
python FieldTiltScan/field_scan_tilt.py --label demo_live --correction hybrid
--x -50 50 3 --y -50 50 3 --z -650 -650 1
```

It parks at (0, 0, −650) and zeroes the gyro (keep the table still), then prints a line per point. Ctrl+C stops it and keeps the points so far. The robot stays enabled at the end, because disabling drops the arms. The scan saves to `FieldTiltScan/data/demo_live_<date>_<time>.csv` ; then run section 5 of `demo_plots.m` , which plots the newest one (heat map and tilt map, raw field about 0.7 G).

| Part of the command | Meaning |
|---|---|
| `--label demo_live` | File name; section 5 looks for `demo_live` |
| `--correction hybrid` | Adjusts each move to cancel the robot's known positioning errors, as in all saved results (needs numpy) |
| `--x MIN MAX N` | x range in mm and number of points, evenly spaced |
| `--y MIN MAX N` | Same for y |
| `--z MIN MAX N` | Height; min = max with 1 point for a single layer |

Keep inside x ±50, y ±150, z −700 to −600 mm (where the correction was validated). Time ≈ points × 13 s.

| Size | Grid options | Points | Time |
|---|---|---|---|
| Quickest | `--x -50 50 2 --y -50 50 2 --z -650 -650 1` | 4 | ~1 min |

| Size | Grid options | Points | Time |
|---|---|---|---|
| Small (demo) | `--x -50 50 3 --y -50 50 3 --z -650 -650 1` | 9 | ~2 min |
| Medium | `--x -50 50 3 --y -100 100 5 --z -650 -650 1` | 15 | ~3–4 min |
| Two layers | `--x -50 50 3 --y -50 50 3 --z -700 -600 2` | 18 | ~4 min |
| Large | `--x -50 50 5 --y -100 100 5 --z -700 -600 2` | 50 | ~11 min |

- At least 2 points in x and y, or the heat map has nothing to fill in.
- `--settle 2` saves about 3 s per point, but the probe may still be swaying.
- Port: the script uses `/dev/cu.usbmodem14101`; check with `ls /dev/cu.usbmodem*` and add `--port /dev/cu.usbmodemXXXX` if it differs.
- Dry run without hardware: `python FieldTiltScan/field_scan_tilt.py --simulate --label demo_test`, then delete the `demo_test` files.
- Before the real demo, delete test scans: `rm /Users/rochelle/FInal/FieldTiltScan/data/demo_live_*`.

## 5b. Live repeatability (two runs back to back)

Two 9-point scans with the magnet fixed take about 4 min; the labels must be exactly `demo_rep1` and `demo_rep2` for section 5b to find them.

Put the magnet under the grid on its stand, and don't touch it, the robot or the table between runs. Both runs in one line (the second starts only if the first finished):

```
cd /Users/rochelle/FInal
python FieldTiltScan/field_scan_tilt.py --label demo_rep1 --correction hybrid
--x -50 50 3 --y -50 50 3 --z -650 -650 1 && python
FieldTiltScan/field_scan_tilt.py --label demo_rep2 --correction hybrid --x
-50 50 3 --y -50 50 3 --z -650 -650 1
```

Or one at a time, to talk between them:

```
python FieldTiltScan/field_scan_tilt.py --label demo_rep1 --correction hybrid
--x -50 50 3 --y -50 50 3 --z -650 -650 1
python FieldTiltScan/field_scan_tilt.py --label demo_rep2 --correction hybrid
--x -50 50 3 --y -50 50 3 --z -650 -650 1
```

- Files: `FieldTiltScan/data/demo_rep1_<date>_<time>.csv` and `demo_rep2_<date>_<time>.csv`. Section 5b takes the newest of each, so older test runs don't matter. Adding `--magnet under` still matches.
- Both runs must use the same `--x --y --z`, or the points won't match.

- Check before plotting: `ls /Users/rochelle/FInal/FieldTiltScan/data/demo_rep*`

Then run section 5b of `demo_plots.m`:

1. Heat map of run 1
2. Heat map of run 2 (each has its own colour scale; read the numbers off the change map)
3. Change from run 1 to run 2 in mG at every point, plus a summary (red = stronger in run 2, blue = weaker, white = no change)

**Say:** the change map should be nearly white. The 6 Oct test changed by 1.4–1.9 mG median per point, about 10× the 0.2 mG sensor noise, so what's left is positioning and settling (about 2 mm). A magnet closer than in that test gives a steeper gradient, so the same position error shows as more mG. Keep the magnet in for both runs: without it the field barely changes across the grid and the test can't see position errors.

## Backups, questions and fixes

Add `"Save", true` to the commands in sections 1–3 the day before, so PNGs sit next to the CSVs if MATLAB is slow on the day.

| Likely question | Answer |
|---|---|
| Why are the x moves off in the dipole test? | The x grid is only 100 mm wide (y is 300 mm), so x is pinned less well; the errors are 1–2σ. A wider x grid would fix it. |
| What's the drift? | About 11 mG over the hour, the same at every point. About 17 % from the probe tilting as it warms, the rest most likely thermal (sensor offset, servo magnets). Take the background close in time, or fit a per-run offset. |
| Where's the Helmholtz field? | All scans so far are coils off; the magnet stands in as a known field source. |
| Is the spread from the sensor? | No: one point's sensor noise is about 0.2 mG and the run-to-run spread about 2 mG, so it comes from positioning and settling. |
| What does `hybrid` mean? | A position correction fitted from the protractor and pen-and-ruler tests that adjusts each move before it's sent. |

| Problem | Fix |
| --- | --- |
| "cd into FieldTiltScan first" | `cd /Users/rochelle/FInal/FieldTiltScan` |
| "Cannot find data/…csv" | MATLAB is in the wrong folder; same fix as above |
| "Not enough input arguments" | A FieldPlots function was run with no inputs; run sections of `demo_plots.m` instead |
| "Undefined function plot_field_layers" | Run section 0 first |
| "No demo_live scan in data/ yet" | Run the live scan in Terminal first |
| "Need a demo_rep1 and a demo_rep2 scan" | Run both back-to-back scans with those exact labels |
| Robot resets or stops mid-move | AA battery pack sagging; fit fresh batteries |